TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Architects at CHI

Citation for published version (APA):
Gamboa, M., Andersen, K., Krogh, P. G., Nair, S., Haghighi, N., & Koulidou, N. (2026). Architects at CHI. In N. Oliver, D. A. Shamma, H. Candello, P. Cesar, P. Lopes, V. Artizzu, F. Draxler, G. Lopez, A. V. Reinschluessel, X. Tong, & P. O. Toups Dugas (Eds.), *CHI EA '26: Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems* Article 793 Association for Computing Machinery, Inc.. https://doi.org/10.1145/3772363.3778785

Document license:
CC BY

DOI:
10.1145/3772363.3778785

Document status and date:
Published: 13/04/2026

Document Version:
Publisher's PDF, also known as Version of Record (includes final page, issue and volume numbers)

Please check the document version of this publication:

• A submitted manuscript is the version of the article upon submission and before peer-review. There can be important differences between the submitted version and the official published version of record. People interested in the research are advised to contact the author for the final version of the publication, or visit the DOI to the publisher's website.
• The final author version and the galley proof are versions of the publication after peer review.
• The final published version features the final layout of the paper including the volume, issue and page numbers.

Link to publication

General rights
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

• Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
• You may not further distribute the material or use it for any profit-making activity or commercial gain
• You may freely distribute the URL identifying the publication in the public portal.

If the publication is distributed under the terms of Article 25fa of the Dutch Copyright Act, indicated by the "Taverne" license above, please follow below link for the End User Agreement:

www.tue.nl/taverne

Take down policy
If you believe that this document breaches copyright please contact us at:

openaccess@tue.nl

providing details and we will investigate your claim.

Download date: 07. Nov. 2026

Architects at CHI

Mafalda Gamboa
Interaction Design, CSE
Chalmers University of Technology, University of Gothenburg
Gothenburg, Sweden
mafalda.gamboa@chalmers.se

Kristina Andersen
Industrial Design
Eindhoven University of Technology
Eindhoven, —, Netherlands
h.k.g.andersen@tue.nl

Peter Gall Krogh
Digital Design
Aarhus University
Aarhus, Denmark
pkrogh@cc.au.dk

Siddharth Nair
Umeå University
Umeå, Sweden
siddharth.nair@umu.se

Nava Haghighi
Computer Science
Stanford University
Stanford, California, USA
navaha@stanford.edu

Nantia Koulidou
Sheffield Creative Industries Institute
Sheffield Hallam University
Sheffield, United Kingdom
n.koulidou@shu.ac.uk

Abstract

This meetup brings together architects at CHI to reflect on how architectural training and practice have shaped trajectories in HCI, and to explore how architectural sensibilities have informed—and could further enrich—the field. We will begin with brief personal narratives: how participants came into HCI or interaction design, what architectural skills and ways of seeing they brought, and which theories continue to guide their work. In a collective mapping exercise, we will examine the relevance of architectural thinking for HCI, surfacing shared insights, divergent practices, and potential avenues for future research. Beyond networking, the session aims to provoke reflection on how architecture has influenced HCI in design processes, pedagogy, and aesthetics, particularly regarding space, materiality, poetics, and human experience. We will also consider critical perspectives, identifying architectural sensibilities that support or challenge current practices, while highlighting canonical architectural theories with promise for informing HCI research and design.

Keywords

architects, architectural theory, architectural sensibilities

ACM Reference Format:
Mafalda Gamboa, Kristina Andersen, Peter Gall Krogh, Siddharth Nair, Nava Haghighi, and Nantia Koulidou. 2026. Architects at CHI. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26), April 13–17, 2026, Barcelona, Spain.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3772363.3778785

1 Goal of the Meet-Up

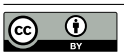
The goal of this CHI Meet-Up is to bring together researchers and practitioners with backgrounds in architecture to explore how architectural training and sensibilities influence our work in human-computer interaction (HCI) and interaction design. Many in the CHI community come from diverse disciplines, but architecture is a particularly rich source of nuance in design knowledge: it brings unique ways of thinking about space, form, materiality, movement, temporality, scale, and sensory experience. With the growing importance of spatial computing, embodied interaction, ubiquitous computing, and immersive environments, architectural perspectives are more relevant than ever. This Meet-Up will create a forum for reflection, networking, and synthesise how architectural ways of thinking can inform, challenge, and enrich HCI research and practice.

2 Background

Human–Computer Interaction (HCI) has repeatedly turned to architecture and related spatial theories as conceptual foundations. A major influence comes from Christopher Alexander's *A Pattern Language* [2] and *The Timeless Way of Building* [1]. Alexander proposed that architectural design knowledge can be captured as reusable patterns that address recurring problems in the built environment. This idea resonated strongly with interaction design, leading to the development of design pattern collections in HCI [4], as well as critical reflections on their adoption and use [5]. Alexander's later work on the theory of living structure and the "quality without a name" has also inspired recent explorations of aesthetics, calm technology, and humane interfaces [12].

HCI has also drawn from phenomenology in architecture, especially through the work of Norberg-Schulz[21], Pallasmaa[24], and Holl [13], which emphasizes lived experience, atmosphere, and embodiment in spatial environments, influencing research on embodied interaction [6] and the role of place and meaning in interaction design [11].

Many HCI authors have made more or less general connections to architecture in their work, for example when showing differences in conceptualisation of interaction [16], or reasoning about human-building interaction [20]. But we want to move towards the recognition of architectural as a theoretical framing, beyond smart buildings. Design sensibilities stemming from architecture have long influenced interaction design. Researchers with architectural training often draw on canonical texts and concepts to inform embodied, material, and experiential aspects of interaction. For example, Tanizaki's *In Praise of Shadows* has been used to cultivate a shared somaesthetic appreciation and sensibility [7], while notions such as inside–outside and Zumthor's *Atmospheres* have



This work is licensed under a Creative Commons Attribution 4.0 International License.
CHI EA '26, Barcelona, Spain
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
https://doi.org/10.1145/3772363.3778785

shape IoT designs that foreground materiality, form, and embodied experience [17, 18]. Likewise, Kinch and Fogtmann describe how architectural education shapes their worldview and informs approaches to interaction design centered on atmospheres [14]. Pallasmaa's critique of CAD has been used to advocate for design practices that preserve material engagement [23]. Krogh and Koskinen's book *Drifting*, written from an architectural perspective, aligns closely with many of the concepts discussed here [19]. Similarly, Andersen and Wakkary's architectural training underpins the theoretical framework of their workshops [3]. Collectively, these works illustrate how architectural ways of seeing, sensing, and structuring space continue to enrich HCI and interaction design.

Apart from direct influences from architecture, there are also concepts that are read differently in HCI and architecture. In this meet-up, we suggest opening a space for agreement and disagreement on the use of such terms and theories. Although originated in anthropology and intercultural communication, one of the most significant is **proxemics**, introduced by Hall in his seminal work *The Hidden Dimension* [10]. Proxemics analyses how people use interpersonal distance, orientation, and spatial zones in communication, and has been extensively adapted in HCI to understand co-located interaction, embodied interfaces, and ubiquitous computing scenarios [9]. In architecture however, this idea leans less on metrics and more on experienced qualities of personal and public space. The concept of affordances, while originally formulated by Gibson in his ecological approach to perception [8], and popularised in design by Norman [22], has strong connections to architectural thinking about how spaces and forms suggest possible actions. In HCI, affordances have become foundational for explaining how interfaces and environments communicate possibilities for interaction, extending architectural concerns with form, materiality, and human action.

3 Activity Overview

We will structure the 90-minute session to maximize participation and reflection. This meetup is focused on identifying and articulating architectural sensibilities, but also to allow participants to network with each other. We plan to do a very simple hands-on activity including collage, while also facilitating conversations on what theories and concepts from architecture the participants have carried with them into HCI.

4 Organisers and Suitability

The Meet-Up will be organised by a set of HCI researchers with varying backgrounds in architecture, but also different origins, paths, and years of presence in the community. We bring experience from academia and practice, and have previously organised design-oriented workshops and discussions at international venues, including many at CHI:

Mafalda Gamboa is a design researcher and PhD Student at Chalmers University of Technology. She studied a bachelors and master's degree in Architecture from Lisbon University. Since then, she worked in a number of architecture offices in Sweden, finally leaving the field in 2018 to keep her own practice. In 2020 she started her PhD in human-computer interaction, focusing on design sensibilities.

Kristina Andersen studied architecture and industrial design at the Royal Danish Academy of Fine Arts in Copenhagen and the Bartlett School of Architecture, before moving to interaction design, working both in industry and in institutions such as Ivrea and STEIM. She is currently associate professor at the Department of Industrial Design at Eindhoven University of Technology (TU/e).

Siddharth Nair is a design researcher and PhD candidate at Umeå University. He trained as an architect and urbanist in South Asia, followed by a master's in design that led to a move to HCI. Since then, he has worked with emerging technologies for social and rural development, and more recently in health and medical contexts, at research spaces across industry and academia.

Peter Gall Krogh is trained as an architect, furniture and product designer, educated from Aarhus School of Architecture, Denmark where he also served as professor until 2016. His professional merits include among others high-end furniture design, consumer products for IKEA and the interior design of Copenhagen suburbian trains (s-tog). He is currently professor at the Department of Digital Design and Information Studies at Aarhus University.

Nava Haghighi is a PhD candidate in computer science at Stanford University. She holds a dual master of science in computer science and integrated design and management from MIT, and a bachelor of architecture from Cal Poly San Luis Obispo. She has worked across urban design and architecture with internationally recognized firms, public art and interactive installations, and experience design, before starting her PhD in 2020. Most recently, she has worked as a researcher with the Human-Centered Machine Intelligence team at Apple.

Nantia Koulidou holds a diploma in architecture from Aristotle University in Greece (2004-2011) and an MSc in Product Design and Information Technology (2012-14). Her practice-led design research is centred around poetic ways of connecting with digital data through objects and jewellery. She often draws on architecture theories and methodologies to bring novel ways of working in HCI. She is currently a senior lecturer at Sheffield Institute of Creative Industries, UK, where she teaches across jewellery and architecture.

5 Community of Interest

This Meet-Up will be of interest to: (a) HCI researchers and practitioners with architectural training; (b) designers and interaction design professionals who work with spatial, environmental, or material concerns; (c) scholars of design research exploring interdisciplinary methods; (d) Students transitioning from architecture into HCI. By providing a dedicated space for architects at CHI, we aim to surface overlooked design sensibilities, foster new collaborations, create a collection of inspirational architectural theories, and take steps in building a lasting community of interest within SIGCHI.

References

[1] Christopher Alexander. 1979. *The Timeless Way of Building.* Oxford University Press.

[2] Christopher Alexander, Sara Ishikawa, and Murray Silverstein. 1977. *A Pattern Language: Towns, Buildings, Construction.* Oxford University Press.

[3] Kristina Andersen and Ron Wakkary. 2019. The Magic Machine Workshops: Making Personal Design Knowledge. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) *(CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300342

[4] Jan O. Borchers. 2000. A pattern approach to interaction design. In *Proceedings of the Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques*. ACM, 369–378.

[5] Andy Dearden and Janet Finlay. 2013. Pattern languages in HCI: A critical review. In *CHI'13 Extended Abstracts on Human Factors in Computing Systems*. ACM, 1989–1998.

[6] Paul Dourish. 2001. *Where the Action Is: The Foundations of Embodied Interaction*. MIT Press, Cambridge, MA.

[7] Mafalda Gamboa and Sjoerd Hendriks. 2024. In Praise of Shadows: Sensibility and Somaesthetic Appreciation for Shadows in Interaction Design. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) *(DIS '24)*. Association for Computing Machinery, New York, NY, USA, 3272–3286. doi:10.1145/3643834.3660679

[8] James J. Gibson. 1979. *The Ecological Approach to Visual Perception*. Houghton Mifflin.

[9] Saul Greenberg, Nicolai Marquardt, Till Ballendat, Robert Diaz-Marino, and Miaosen Wang. 2013. Proxemics in human-computer interaction. In *Proceedings of the 2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. ACM, 285–294.

[10] Edward T. Hall. 1966. *The Hidden Dimension*. Anchor Books.

[11] Steve Harrison, Deborah Tatar, and Phoebe Sengers. 2007. The three paradigms of HCI. In *Proceedings of CHI 2007 Workshop on Perspectives on HCI Research*.

[12] Marc Hassenzahl, Kerstin Eckoldt, and Volker Wulf. 2023. Bringing design patterns to life: Applying Christopher Alexander's theory of living structure in HCI. *International Journal of Human–Computer Interaction* 39, 13 (2023), 2451–2468.

[13] Steven Holl, Juhani Pallasmaa, and Alberto Pérez-Gómez. 1994. *Questions of Perception: Phenomenology of Architecture*. William Stout Publishers, San Francisco. Expanded edition published 2006.

[14] Sofie Kinch. 2014. *Designing for Atmospheric Experiences: Taking an Architectural Approach to Interaction Design: Industrial PhD Dissertation*. Aarhus School of Architecture.

[15] Sofie Kinch and Majken Hillerup Fogtmann. 2011. The Design Researcher in Quandary: The Conflicting Roles of Being Both a Designer and a Design Researcher. In *Proceedings of the Second Conference on Creativity and Innovation in Design (DESIRE '11)*. ACM, Eindhoven, The Netherlands, 193–196. doi:10.1145/2079216.2079244

[16] David Kirsh. 2019. Do Architects and Designers Think about Interactivity Differently? *ACM Trans. Comput.-Hum. Interact.* 26, 2, Article 7 (April 2019), 43 pages. doi:10.1145/3301425

[17] Nantia Koulidou, Luís P. Carvalho, and Daniela Petrelli. 2024. Art Digital Jewellery as Atmospheres: An autobiographical RtD exploration into IoT for poetic contexts. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) *(DIS '24)*. Association for Computing Machinery, New York, NY, USA, 2374–2387. doi:10.1145/3643834.3660734

[18] Niki Koulidou and Jayne Wallace. 2017. Staged Atmosphere: The Air[craft] Workshop. In *Proceedings of the 3rd Biennial Research Through Design Conference*. Edinburgh, UK, 372–388. doi:10.6084/m9.figshare.4747003 Article 24.

[19] Peter Gall Krogh and Ilpo Koskinen. 2020. *Drifting by Intention: Four Epistemic Traditions from within Constructive Design Research*. Springer, Cham. doi:10.1007/978-3-030-37896-7

[20] Eleni Margariti, Vasilis Vlachokyriakos, and David Kirk. 2023. Understanding occupants' experiences in quantified buildings: results from a series of exploratory studies.. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 640, 15 pages. doi:10.1145/3544548.3581256

[21] Christian Norberg-Schulz. 1980. *Genius Loci: Towards a Phenomenology of Architecture*. Rizzoli, New York.

[22] Donald A. Norman. 1988. *The Psychology of Everyday Things*. Basic Books.

[23] Martin A. Padalak and Tom Jenkins. 2022. Being in the world with the aural-visualiser 1000. In *Nordic Human-Computer Interaction Conference* (Aarhus, Denmark) *(NordiCHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 89, 1 pages. doi:10.1145/3546155.3547281

[24] Juhani Pallasmaa. 2024. *The Eyes of the Skin: Architecture and the Senses* (4th ed.). John Wiley & Sons.